

Multiple Choice (10 marks)

1. The rotational spectrum of HF has lines 41.9 cm^{-1} apart. Calculate the moment of inertia and the bond length.

2. $1.256 \times 10^{-46}\text{ kg m}^2$ and $9.100 \times 10^{-2}\text{ nm}$

2. $1.336 \times 10^{-47}\text{ kg m}^2$ and $8.250 \times 10^{-2}\text{ nm}$

2. $1.336 \times 10^{-46}\text{ kg m}^2$ and $9.196 \times 10^{-1}\text{ nm}$

2. $1.410 \times 10^{-47}\text{ kg m}^2$ and $9.250 \times 10^{-2}\text{ nm}$

2. $1.256 \times 10^{-47}\text{ kg m}^2$ and $8.196 \times 10^{-2}\text{ nm}$

Calculate the energy of a photon for a wavelength of 100 pm (about one atomic diameter). \$

(a) $4 \times 10^{-15}\text{ J}$

(a) $1.2 \times 10^{-15}\text{ J}$

(a) $5 \times 10^{-15}\text{ J}$

(a) $3.5 \times 10^{-15}\text{ J}$

(a) $2 \times 10^{-15}\text{ J}$

Less than 1/1000 of the mass of any atom is contributed by

(a) the protons, electrons and neutrons

(b) the electrons and protons

(c) the nucleus

(d) the protons

(e) the neutrons

What is the electromotive force (emf) of a system when K, the equilibrium constant for the reaction = Q, the concentration quotient?

(a) -2

(b) 1

(c) -0.5

(d) 0.5

(e) 2

What is the heat capacity of uranium? Its atomic heat is 26 J/mole and its heat capacity is 238 g/mole.

- (a) 0.11 J/g
- (a) 0.20 J/g
- (a) 0.13 J/g
- (a) 0.18 J/g
- (a) 0.30 J/g

True / False (10 marks)

Answer True or False

- 6. True or False: The oxidation number of chlorine in the perchlorate ion, ClO_4^- , is +5.
- 7. True or False: Ammonia, NH_3 , is the strongest base in liquid ammonia.
- 8. True or False: An atom of ^{52}Cr contains 24 protons, 28 neutrons, and 24 electrons.
- 9. True or False: Compounds formed from Sn, P, and Si include SnF_2 , PH_5 , and SiO_4 , which are incorrect based on valence rules.
- 10. True or False: The linewidth of an NMR line with a T_2 of 15 ms is calculated to be approximately 30.1 Hz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the reasons why the ^1H NMR spectrum of chloroform (CHCl_3) appears as a singlet, considering the effects of halogen isotopes and their electric quadrupole moments.
- 12. Given an unknown element X with an atomic weight of 210.197 amu and two isotopes (^{210}X at 209.64 amu and ^{212}X at 211.66 amu, calculate the relative abundance of each isotope and discuss the

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